AI Assignment Submission

Problem Statement 0-B: Travelling Agent

Informed Search (IDA*) and Local Search (Hill Climbing)

Course	Artificial and Computational Intelligence
Assignment	Assignment 1
Problem	0-B
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Part A - Theory (PEAS and Method Explanation)

1. Agent Environment and PEAS

Problem Context

A travelling agent must move from Chennai (node A) to Hyderabad (node X) on a weighted graph. Edge weights represent approximate travel time between cities. The task is to identify a valid route to the goal using the required search methods (IDA* and Hill Climbing), while reporting path and complexity.

PEAS Description

- **Performance Measure:** Reach goal node X with minimum travel cost; produce valid path; report time and space complexity.
- **Environment:** Weighted graph of cities, static during one run, fully observable.
- **Actuators:** Move from current node to an adjacent node connected by an edge.
- **Sensors:** Current node, neighboring nodes, and edge costs from graph representation.

Environment Characteristics

- **Fully observable:** all nodes, edges, and costs are known from the start.
- **Deterministic:** moving on an edge always gives the same next node and cost.
- **Sequential:** each decision affects subsequent choices in the path.
- **Static:** topology and weights do not change during execution.
- **Discrete:** nodes and transitions are finite and countable.

2. Heuristic / Fitness Function Definition

For this problem, the heuristic follows the assignment statement exactly:

> `h(n) = average of total travel costs across all simple paths from node n to goal node X.`

Interpretation

- Lower h(n) indicates that node n is generally closer (in travel-time sense) to X.
- The heuristic is computed from graph structure, not manually guessed.

Usage in Algorithms

- **IDA*:** uses f(n) = g(n) + h(n), where g(n) is path cost from start to n.
- **Hill Climbing:** selects neighbor with best (lowest) heuristic value; if no improving neighbor exists, local optimum behavior is reported.

3. Data Structures Used in Implementation

- **Graph:** Dictionary of dictionaries (adjacency map with edge weights).
- **Path containers:** list for ordered path sequence.
- **Visited tracking:** set to prevent node revisits in simple-path exploration.
- **Heuristic table:** dictionary mapping node to heuristic value h(n).
- **Result object:** structured record containing path, cost, expanded nodes, depth, and flags.

4. Informed and Local Search Methods (Summary)

IDA* (Informed Search)

- Starts with threshold = h(start).
- Performs depth-first bounded search using f(n).
- If goal not found, threshold increases to minimum exceeded f-cost and repeats.
- Returns path A → D → G → X (cost 19) in the submitted implementation run.

Hill Climbing (Local Search)

- At each step chooses best heuristic neighbor.

- Tracks local optimum condition when strict heuristic improvement is not possible.
- Submitted run reports path A → D → G → X and marks local-optimum encountered as Yes.

5. Complexity Reporting Approach

The code prints complexity information explicitly:

- **IDA*:** Time O(b^d) worst case, Space O(bd) worst case.
- **Hill Climbing:** Time O(b) per move, Space O(b) for neighbor inspection.

Part B - Implementation (Code and Runtime Output)

6. Runtime Output (Markdown-Style Code Block)

```
Assignment 1 - Problem Statement 0-B
Graph search from Chennai (A) to Hyderabad (X)

Heuristic values (average cost of all simple paths to X):
  h(A) = 22
  h(B) = 27
  h(C) = 31.3333
  h(D) = 19
  h(E) = 24.5
  h(F) = 21.6667
  h(G) = 20.3333
  h(H) = 31.6667
  h(X) = 0

IDA* result:
  Path: A -> D -> G -> X
  Total cost: 19
  Threshold progression: 22, 27.3333
  Nodes expanded: 5
  Max depth reached: 3
  Time complexity: O(b^d) in the worst case
  Space complexity: O(bd) in the worst case

Hill climbing result:
  Path: A -> D -> G -> X
  Total cost: 19
  Nodes expanded: 3
  Max depth reached: 3
  Reached goal: Yes
  Local optimum encountered: Yes
  Time complexity: O(b) per move
  Space complexity: O(b) for neighbor inspection
```

7. Python Source Code

```
from dataclasses import dataclass
from math import inf
from statistics import fmean
from typing import Dict, Iterable, List, Optional, Set, Tuple

Graph = Dict[str, Dict[str, int]]

GRAPH: Graph = {
    "A": {"B": 5, "D": 3},
    "B": {"A": 5},
    "C": {"G": 11},
    "D": {"A": 3, "E": 12, "F": 6, "G": 4},
    "E": {"D": 12, "F": 7},
    "F": {"D": 6, "E": 7, "H": 10, "X": 8},
    "G": {"C": 11, "D": 4, "X": 12},
    "H": {"F": 10},
    "X": {"F": 8, "G": 12},
}

START_NODE = "A"
GOAL_NODE = "X"

@dataclass
class SearchResult:
    path: List[str]
    cost: float
    nodes_expanded: int
    max_depth: int
    thresholds: Optional[List[float]] = None
    local_optimum: bool = False

def _all_simple_paths_with_costs(graph: Graph, start: str, goal: str) -> List[Tuple[List[str], int]]:
    paths: List[Tuple[List[str], int]] = []
    def dfs(node: str, current_path: List[str], current_cost: int, visited: Set[str]) -> None:
        if node == goal:
            paths.append((current_path.copy(), current_cost))
            return
        for neighbor, weight in graph[node].items():
            if neighbor in visited:
                continue
            visited.add(neighbor)
            current_path.append(neighbor)
            dfs(neighbor, current_path, current_cost + weight, visited)
```

```
            current_path.pop()
            visited.remove(neighbor)
    dfs(start, [start], 0, {start})
    return paths

def build_average_heuristic(graph: Graph, goal: str) -> Dict[str, float]:
    heuristic: Dict[str, float] = {}
    for node in graph:
        paths = _all_simple_paths_with_costs(graph, node, goal)
        heuristic[node] = fmean(cost for _, cost in paths) if paths else inf
    return heuristic

def hill_climbing(graph: Graph, start: str, goal: str, heuristic: Dict[str, float]) -> SearchResult:
    current, path, cost = start, [start], 0.0
    visited = {current}
    nodes_expanded = max_depth = 0
    encountered_local_optimum = False
    while current != goal:
        nodes_expanded += 1
        neighbors = [(heuristic[nb], ecost, nb) for nb, ecost in graph[current].items() if nb not in visited or nb == goal]
        if not neighbors:
            return SearchResult(path, cost, nodes_expanded, max_depth, local_optimum=encountered_local_optimum)
        neighbors.sort()
        improving = [n for n in neighbors if n[0] < heuristic[current]]
        if improving:
            _, edge_cost, best = improving[0]
        else:
            encountered_local_optimum = True
            _, edge_cost, best = neighbors[0]
        if best in visited and best != goal:
            return SearchResult(path, cost, nodes_expanded, max_depth, local_optimum=encountered_local_optimum)
        current, cost = best, cost + edge_cost
        path.append(current)
        visited.add(current)
        max_depth = max(max_depth, len(path) - 1)
    return SearchResult(path, cost, nodes_expanded, max_depth, local_optimum=encountered_local_optimum)

def ida_star(graph: Graph, start: str, goal: str, heuristic: Dict[str, float]) -> SearchResult:
    thresholds, nodes_expanded, max_depth = [], 0, 0
    def search(node: str, g_cost: float, threshold: float, path: List[str], visited: Set[str], depth: int):
        nonlocal nodes_expanded, max_depth
        max_depth = max(max_depth, depth)
        f_cost = g_cost + heuristic[node]
        if f_cost > threshold:
            return None, f_cost
        if node == goal:
            return path.copy(), g_cost
        nodes_expanded += 1
        minimum_excess = inf
        neighbors = sorted(((g_cost + ecost + heuristic[nb], ecost, nb) for nb, ecost in graph[node].items() if nb not in
visited), key=lambda x: (x[0], x[1], x[2]))
        for _, edge_cost, nb in neighbors:
            visited.add(nb)
            path.append(nb)
            found_path, next_bound = search(nb, g_cost + edge_cost, threshold, path, visited, depth + 1)
            path.pop()
            visited.remove(nb)
            if found_path:
                return found_path, sum(graph[path[i]][path[i+1]] for i in range(len(path)-1))
            minimum_excess = min(minimum_excess, next_bound)
        return None, minimum_excess
    threshold = heuristic[start]
    while True:
        thresholds.append(threshold)
        found_path, result = search(start, 0.0, threshold, [start], {start}, 0)
        if found_path:
            return SearchResult(found_path, result, nodes_expanded, max_depth, thresholds)
        if result == inf:
            return SearchResult([], inf, nodes_expanded, max_depth, thresholds)
        threshold = result

heuristic = build_average_heuristic(GRAPH, GOAL_NODE)
ida_result = ida_star(GRAPH, START_NODE, GOAL_NODE, heuristic)
hill_result = hill_climbing(GRAPH, START_NODE, GOAL_NODE, heuristic)

print("Assignment 1 - Problem Statement 0-B")
print("Graph search from Chennai (A) to Hyderabad (X)")
print()
print("Heuristic values (average cost of all simple paths to X):")
for node in sorted(heuristic):
    value = heuristic[node]
    display = "inf" if value == inf else f"{value:.4f}".rstrip("0").rstrip(".")
    print(f"  h({node}) = {display}")
print()
print("IDA* result:")
print(f"  Path: {' -> '.join(ida_result.path)}")
print(f"  Total cost: {ida_result.cost:.0f}")
print(f"  Threshold progression: {', '.join(f'{v:.4f}'.rstrip('0').rstrip('.') for v in ida_result.thresholds or [])}")
print(f"  Nodes expanded: {ida_result.nodes_expanded}")
print(f"  Max depth reached: {ida_result.max_depth}")
print("  Time complexity: O(b^d) in the worst case")
print("  Space complexity: O(bd) in the worst case")
print()
print("Hill climbing result:")
print(f"  Path: {' -> '.join(hill_result.path)}")
print(f"  Total cost: {hill_result.cost:.0f}")
print(f"  Nodes expanded: {hill_result.nodes_expanded}")
print(f"  Max depth reached: {hill_result.max_depth}")
print(f"  Reached goal: {'Yes' if hill_result.path and hill_result.path[-1] == GOAL_NODE else 'No'}")
print(f"  Local optimum encountered: {'Yes' if hill_result.local_optimum else 'No'}")
```

```
print("  Time complexity: O(b) per move")
print("  Space complexity: O(b) for neighbor inspection")
```